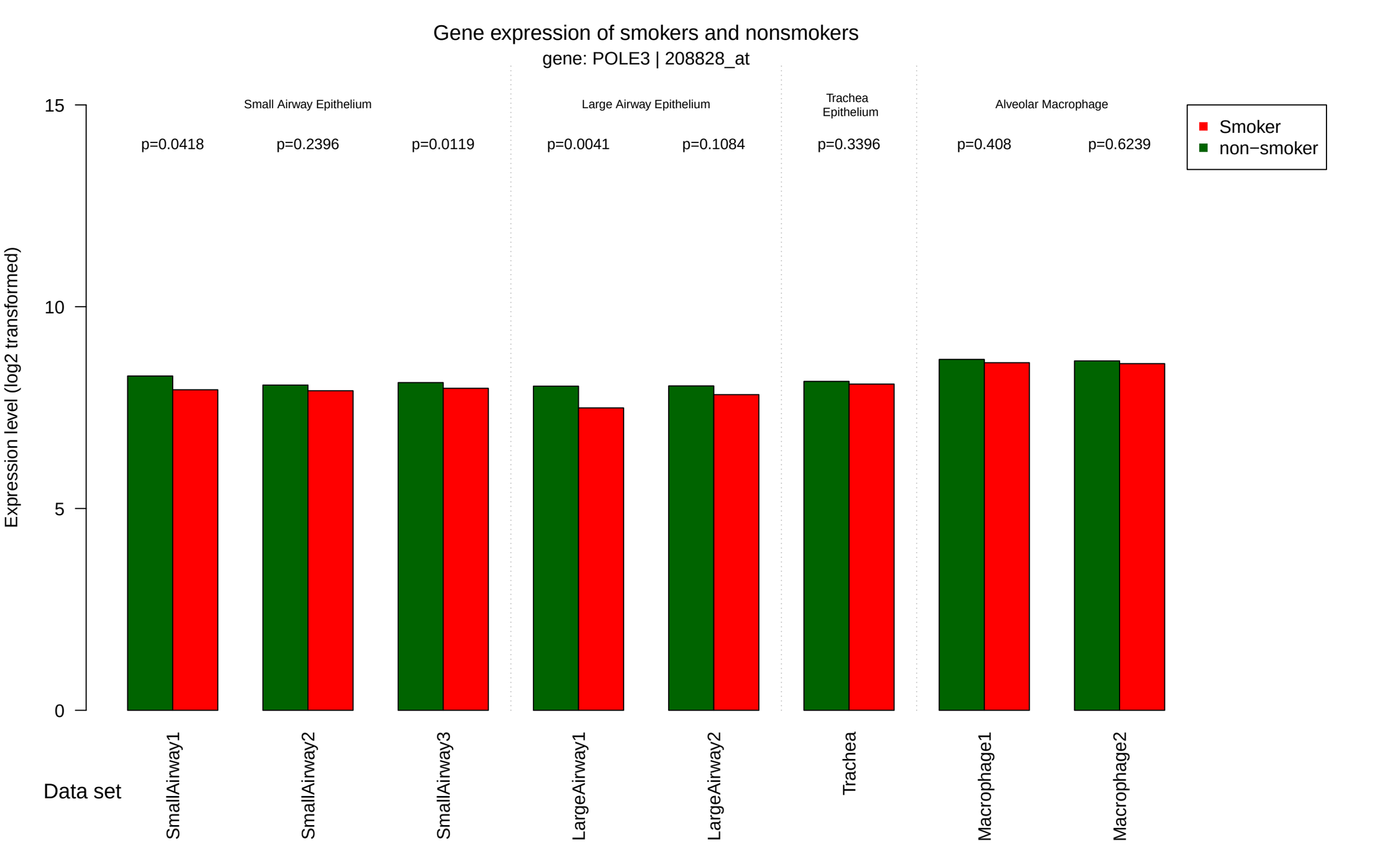
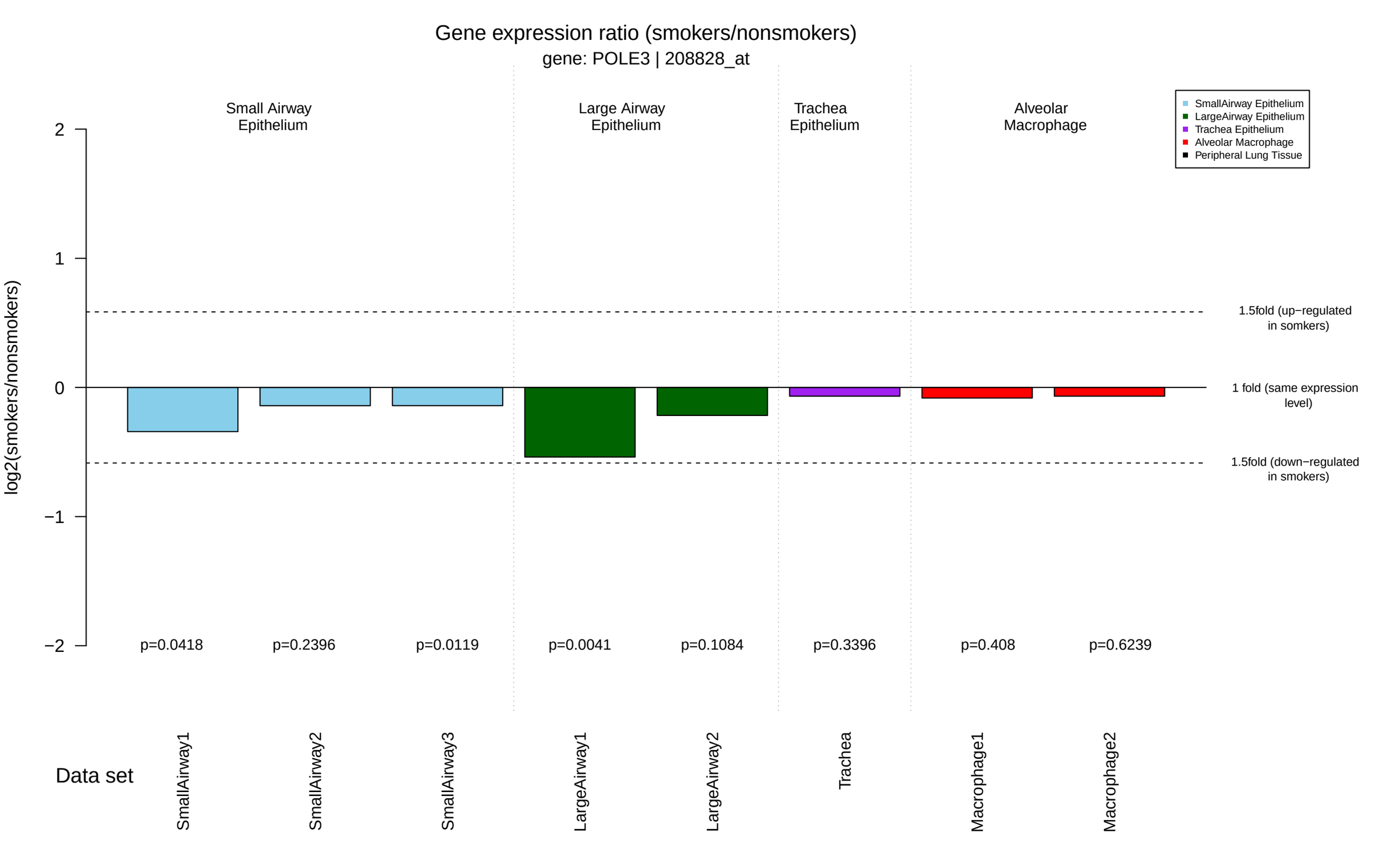
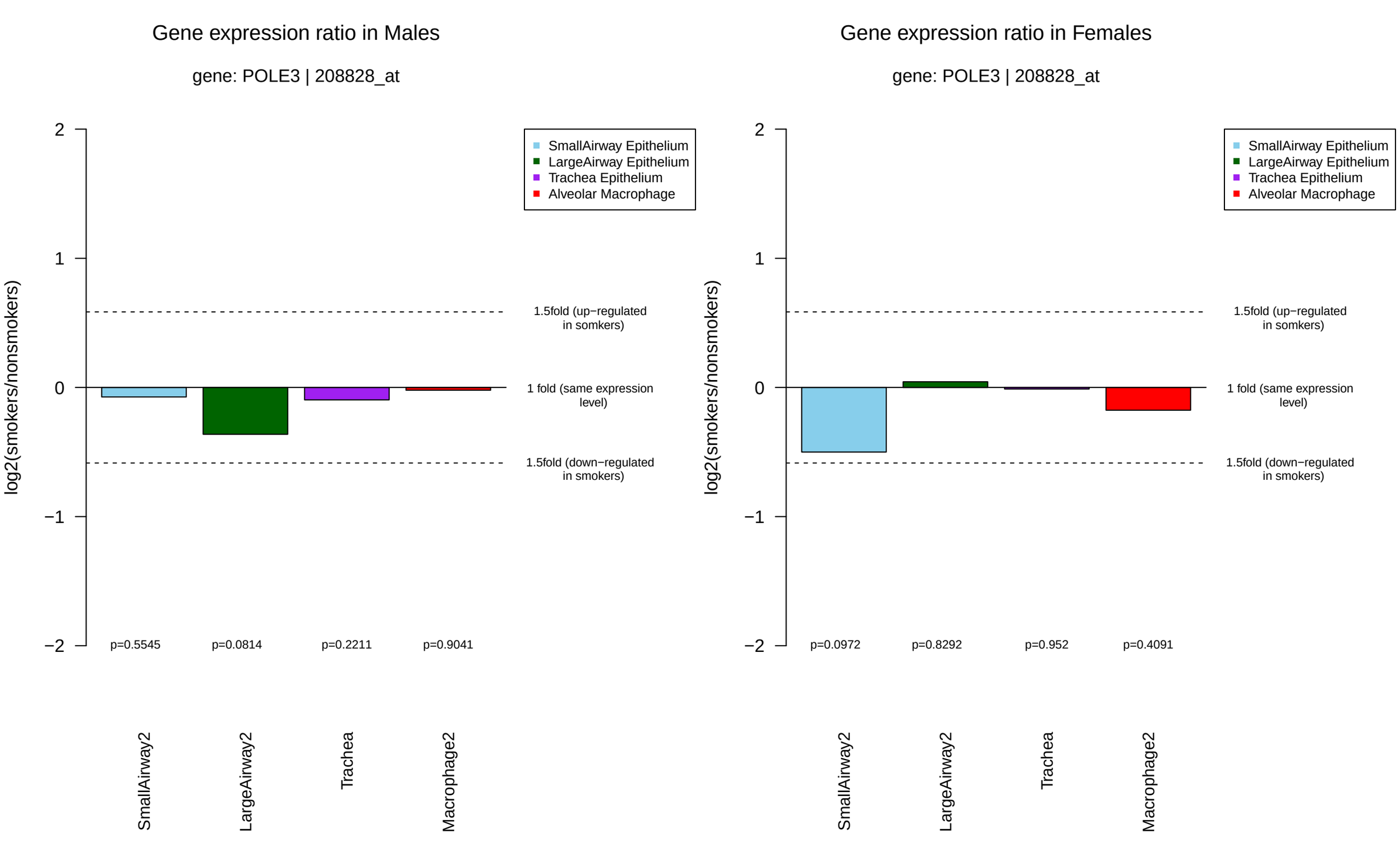
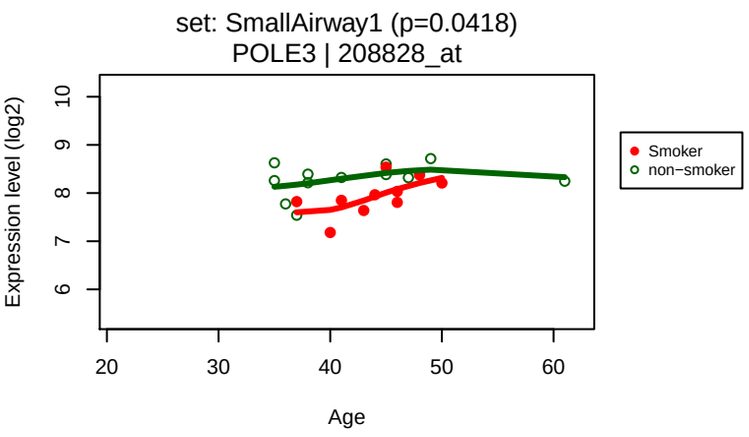
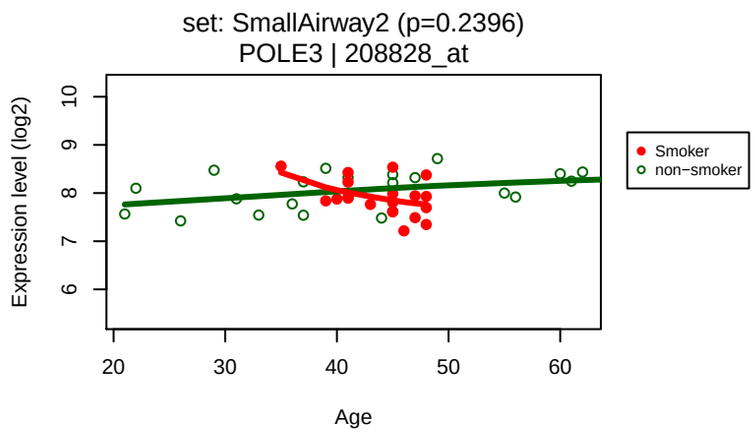
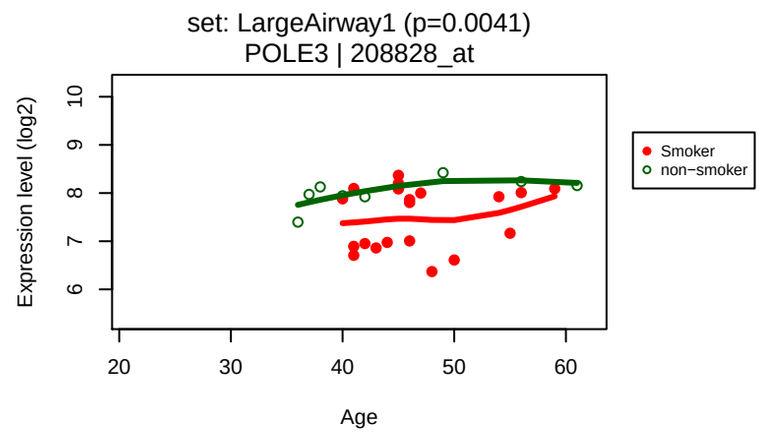

Gene expression of smokers and nonsmokers
gene: POLE3 | 208828_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0418
p=0.2396
p=0.0119
p=0.0041
p=0.1084
p=0.3396
p=0.408
p=0.6239
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

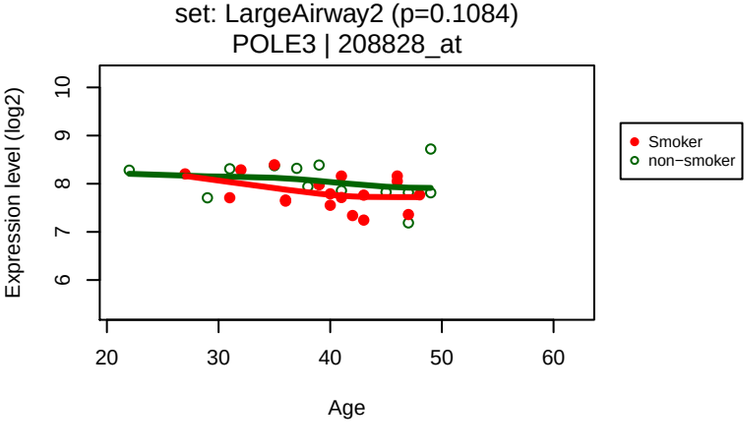

Gene expression ratio (smokers/nonsmokers)
gene: POLE3 | 208828_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0418
p=0.2396
p=0.0119
p=0.0041
p=0.1084
p=0.3396
p=0.408
p=0.6239
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

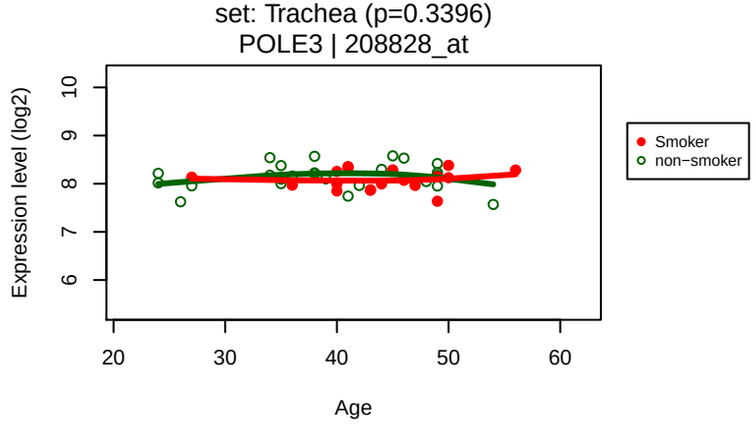

Gene expression ratio in Males
gene: POLE3 | 208828_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5545
p=0.0814
p=0.2211
p=0.9041
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLE3 | 208828_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0972
p=0.8292
p=0.952
p=0.4091
SmallAirway2
LargeAirway2
Trachea
Macrophage2

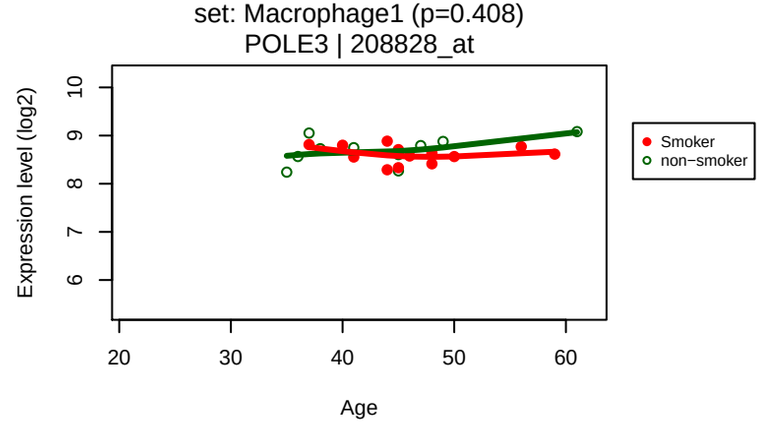

set: SmallAirway1 (p=0.0418)
POLE3 | 208828_at
Expression level (log2)
Age
Smoker
non−smoker

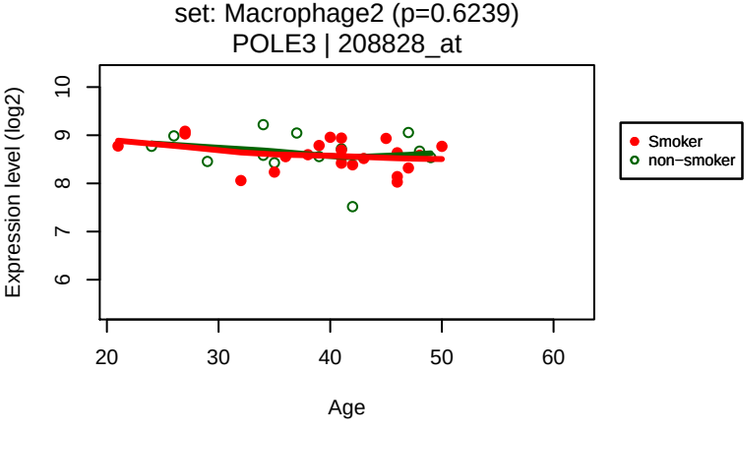

set: SmallAirway2 (p=0.2396)
POLE3 | 208828_at
Expression level (log2)
Age
Smoker
non−smoker

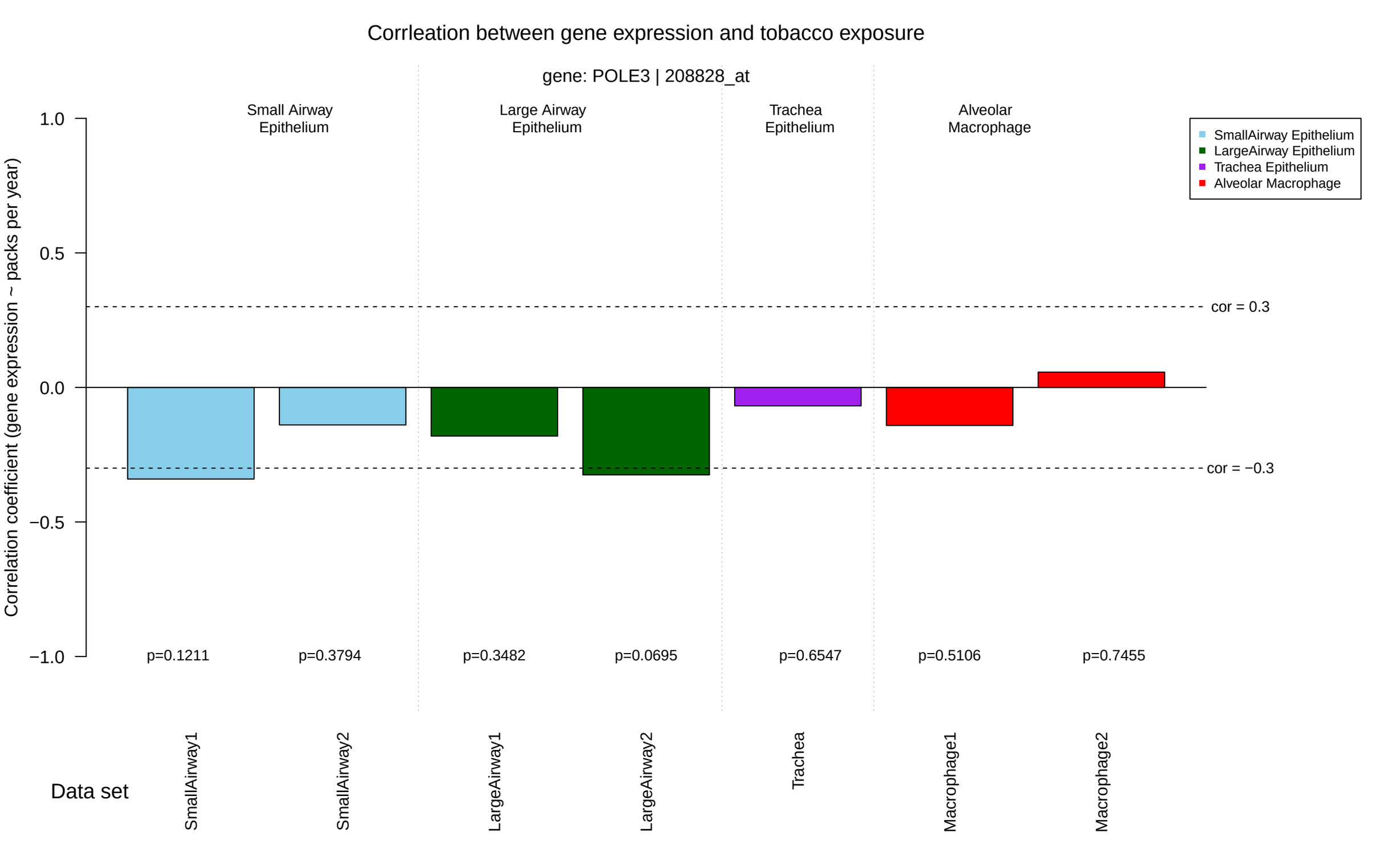

set: LargeAirway1 (p=0.0041)
POLE3 | 208828_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1084)
POLE3 | 208828_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3396)
POLE3 | 208828_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.408)
POLE3 | 208828_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6239)
POLE3 | 208828_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLE3 | 208828_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1211
p=0.3794
p=0.3482
p=0.0695
p=0.6547
p=0.5106
p=0.7455
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2